

Employment Protection Deregulation and Deaths of Despair

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Abstract

Income and wealth inequality have risen sharply within advanced economies, raising concerns about their social and health consequences. While the link between inequality and adverse health outcomes remains debated, the phenomenon of “deaths of despair”—drug overdoses, alcohol-related diseases, and suicide—has drawn attention in the United States. This study extends the analysis globally by examining whether labor market deregulation, specifically employment protection legislation (EPL) reforms that weaken workers’ bargaining power, contributes to deaths of despair. Using OECD indicators of employment protection and WHO mortality data for 26 advanced economies from 1998 to 2019, we estimate the dynamic effects of major EPL reforms on age-standardized death rates. Applying Jordà’s (2005) local projection method, we trace the temporal response of mortality to deregulation. Results show a significant rise in deaths from drug use disorders and suicide within five years of major EPL reforms. Further analysis indicates that governmental social transfers, unemployment benefit replacement rates, public health coverage, avoidable mortality, and poverty levels mediate these effects. These findings highlight the health risks of labor market deregulation and underscore the need for policies that strengthen social safety nets to mitigate deaths of despair.

Keywords: Deaths of Despair; Employment Protection Legislation; Labor Market Deregulation; Mortality; Social Safety Nets; OECD Economies

JEL Classification Numbers: I14; J65; J08; I18

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1 Introduction

Over the past three decades, advanced economies have experienced profound changes in the organization of work and the distribution of economic gains. A salient trend is the steady weakening of workers’ bargaining power and job security, often attributed to deregulation of employment protection legislation (EPL), declining union density, and the rise of non-standard work. In parallel, researchers and policy-makers have grown increasingly concerned about “deaths of despair”—mortality from drug overdoses, alcohol-related diseases, and suicide—which have contributed to stalled or reversed improvements in life expectancy in several high-income countries. These twin developments raise an urgent question: do labor-market institutions that shape job security and income stability have measurable consequences for despair-related mortality?

This paper examines that question in a cross-national setting by studying the dynamic health effects of major EPL deregulation. We assemble annual data for 26 OECD economies from 1998 to 2019 and focus on age-standardized death rates for drug use disorders, alcohol use disorders, and suicide. To trace the timing and persistence of effects—while avoiding strong parametric assumptions—we use local projections (Jordà, 2005) to estimate impulse-response functions (IRFs) following major reforms to EPL that relax dismissal protections for workers on regular contracts and alter procedures for collective dismissals. This approach allows us to characterize how deaths of despair evolve in the years immediately after reforms and to differentiate medium-run trajectories across causes of death.

Our analysis yields three core findings. First, major EPL deregulation is followed by a statistically and economically significant rise in drug-related mortality, with effects that build over several years; we also find a positive and significant increase in suicide mortality within five years of reform. Second, the average effect on alcohol-related mortality is not statistically different from zero, pointing to heterogeneous causal pathways across the components of deaths of despair. Third, we detect important institutional moderators: countries with more generous unemployment insurance, larger social transfers, stronger public health-insurance coverage, and more effective health systems (lower avoidable mortality) exhibit attenuated adverse responses, consistent with an institutional-buffering mechanism that dampens the translation of employment shocks into fatal outcomes. We further document increases in violence and inflammatory heart disease mortality several years after deregulation, patterns plausibly linked to the direct and indirect harms of substance use as well as to prolonged socioeconomic stress.

These results contribute to several literatures. In labor economics, a large body of work evaluates the efficiency and distributional effects of job-protection reforms, yet far fewer studies follow their downstream health consequences. By linking major EPL changes to cause-specific mortality dynamics, we provide new evidence that labor-market flexibility carries measurable population-health trade-offs. In population health and social epidemiology, the paper complements individual-level studies connecting insecurity, job loss, and precarious work to mental health, substance use, and suicide by showing that macro-institutional shifts can move despair-related mortality at the country level. Finally, by disaggregating deaths of despair and emphasizing timing, magnitude, and moderators, we speak to debates about whether rising mortality reflects supply-side shocks (e.g., synthetic opioids) or deeper structural distress: our results are most consistent with a view in which institutional erosion increases exposure and vulnerability, particularly where social protections are thin, even as drug supply conditions shape which harms are most salient.

Methodologically, the paper advances a design that is well-suited to policy shocks that occur at different times across countries. We identify major EPL reforms using large discrete movements in the OECD’s EPL indicator and embed those events in a panel local-projection framework with country and year fixed effects, forward-looking reform controls, and lags of outcomes and treatments to mitigate reverse causality. We then extend the baseline with interaction terms that capture how country characteristics—unemployment-benefit replacement rates, social-transfer effort, public health-insurance

coverage, avoidable mortality, and poverty—amplify or dampen the mortality response. Throughout, we focus on age-standardized death rates to ensure comparability across countries and leverage the ICD-10 era to avoid spurious discontinuities associated with ICD transitions.

The findings have direct policy relevance. If reforms that loosen dismissal protections increase deaths of despair—especially drug and suicide mortality—then labor-market institutions should be considered part of the public-health architecture. The moderation we observe in countries with stronger income supports and health systems suggests that policymakers contemplating flexibility-enhancing reforms should pair them with robust social insurance and health-system capacity, including accessible mental-health and addiction services. More broadly, the evidence cautions against evaluations of labor-market reform that focus narrowly on employment or wage aggregates while overlooking externalities borne by health systems, families, and communities.

The remainder of the paper proceeds as follows. Section 2 reviews related literature on EPL deregulation, economic insecurity, and deaths of despair, and motivates our focus on institutional buffering. Section 3 presents a conceptual framework linking deregulation to mortality through economic, psychosocial, and institutional channels. Section 4 describes the data sources, construction of outcomes, and identification of major reforms. Section 5 details the empirical strategy based on local projections and the heterogeneous-effects design. Section 6 reports the main IRFs and moderator analyses and discusses mechanisms. Section 7 concludes with implications for policy and future research.

2 Literature Review

This section reviews the existing literature on employment protection deregulation, labor market institutions, and their relationship to mental health and mortality, particularly deaths of despair. It highlights empirical findings, theoretical debates, and gaps that motivate the current study.

2.1 Employment Protection Deregulation (EPD) and Labor Market Outcomes

The deregulation of employment protection legislation (EPL) has been a central feature of labor market reforms in advanced economies since the 1980s. These reforms aim to enhance labor market flexibility, reduce unemployment, and improve competitiveness. However, their distributional and social consequences remain contested.

Ciminelli et al. (2022) provide one of the most comprehensive empirical assessments of EPL deregulation. Using a newly constructed dataset covering 26 advanced economies from 1970 to 2013, they find that job protection deregulation significantly reduces the labor share of income. Their analysis suggests that such reforms may have contributed to approximately 15% of the average labor share decline in these economies. This finding underscores the need to consider equity-efficiency trade-offs in labor market policy design.

Barbieri and Cutuli (2016) examine the labor market effects of employment protection deregulation in Europe, focusing on “marginal” reforms that relax protection primarily for temporary contracts while leaving permanent employment largely unchanged. Using pseudo-panel data from 13 European countries (1992–2008), they show that such partial deregulation increases labor market dualism without delivering sustained employment gains. In particular, reforms widening the gap between protections for permanent and temporary workers generate at most short-lived (“honeymoon”) increases in employment, but ultimately reduce the availability of stable jobs and do not significantly improve overall employment rates. At the micro level, temporary jobs provide only limited protection against future unemployment and rarely function as stepping stones into stable employment, instead often reinforcing precarious career

trajectories. Overall, the authors conclude that targeted EPL deregulation can exacerbate inequality and employment instability by increasing insider–outsider divides, while failing to achieve meaningful or lasting improvements in labor market performance.

While some studies document the impact on labor markets following EPL deregulation, considerably less attention has been devoted to its broader social consequences. In particular, the downstream effects on health outcomes and mortality remain largely unexplored. This gap is especially notable given growing evidence linking labor market insecurity to adverse health conditions. Reforms that weaken job protection may increase economic uncertainty, precarious employment, and income volatility, all of which are potential risk factors for deteriorating physical and mental health. Yet, systematic evidence on whether and how EPL deregulation contributes to outcomes such as “deaths of despair” remains scarce. Addressing this gap is crucial for a more comprehensive evaluation of labor market reforms that accounts not only for efficiency and employment effects, but also for their broader societal costs.

2.2 Determinants of Deaths of Despair (DoD)

The concept of “deaths of despair,” popularized by Case and Deaton (2020), refers to rising mortality from suicide, drug overdose, and alcohol-related diseases among middle-aged individuals, particularly those without a college degree. Their work links these deaths to long-term economic decline, loss of stable employment, and weakening social institutions.

Having a bachelor’s degree (BA) is argued to be a dominant axis of U.S. midlife mortality. Case and Deaton (2022) shows that sharp increases in deaths of despair are concentrated among individuals without a bachelor’s degree, a pattern not observed in other high-income countries. The ratio of the US population without a BA or higher is 61.3% in 2024, underscoring the broad relevance of this divide (U.S. Census Bureau, 2025). Case and Deaton (2020) link these deaths of despair to worsening economic and social conditions among the less educated, including increasing pain and morbidity, stagnant real wages, weakened attachment to employment, reduced religious participation, increasing nonmarital childbearing, and lower marriage rates. Declining attachment to employment reflects the erosion of stable, meaningful jobs that provide identity and opportunities for advancement. They further highlight the roles of an increasingly costly healthcare system, limited social safety nets for working-age adults, and the adverse effects of meritocratic pressures on those left behind. Case and Deaton (2022) documents rising physical pain, mental distress, and suicide rate as proximate causes of despair.

Ruhm (2018) examines the rapid rise in U.S. drug overdose deaths and evaluates whether it is primarily driven by “deaths of despair”, long-term economic decline and social distress or by changes in the drug environment, such as shifts in drug availability and risk. Ruhm highlights several puzzles not solved by the “deaths of despair” hypothesis, including differences across racial groups and countries that are difficult to reconcile with a purely economic explanation. As an alternative, Ruhm argues that changes in the drug environment, including prescribing practices and drug potency, may better account for observed patterns. The study proposes to test these competing explanations using county-level data by examining whether economic decline predicts increases in drug, suicide, and alcohol-related deaths and whether shifts in the types of drugs driving overdoses correspond to changes in the demographic groups affected. The central idea is that if economic distress is the main cause, the composition of victims should remain stable, whereas if the drug environment is key, the groups most affected should change alongside evolving drug environments. Findings indicate that economic conditions explain only a small portion of the increase in drug mortality, pointing instead to drug-related factors as the primary driver.

A complementary critique emphasizes component heterogeneity of DoD and timing. Ruhm (2022) shows that the post-1999 rise in midlife White mortality is predominantly drug-overdose driven, with suicide and alcohol-related liver disease playing smaller and age-specific roles. Consequently, the omnibus

label “deaths of despair” risks obscuring distinct causal channels and policy levers. He also argues that the late-1990s onset and subsequent surges align more closely with drug-supply shocks including prescription opioids, then heroin substitution, and, after 2013, illicit fentanyl where socioeconomic distress likely shaped who was most exposed rather than when and how much of the mortality rise. These points reinforce a choice to model components of DoD separately and to include explicit supply-side moderators in an empirical model.

Regarding the criticism on the “deaths of despair” hypothesis, Case and Deaton (2022) responded in a way that rather than attributing the crisis to a single or dominant cause, deaths of despair arise from the interaction of multiple factors, including both supply-side and demand-side dynamics. In particular, the opioid epidemic likely would not have reached its current scale without the underlying pain, distress, and deteriorating living conditions experienced by less-educated Americans. Thus, while the expansion of opioid supply may have initiated the crisis, its rapid escalation can be understood as a consequence of demand rooted in despair. This interpretation is further supported by the fact that more educated Americans have been relatively less affected by opioid mortality, suggesting that differences in social and economic resources, rather than exposure to drugs alone, play a crucial role.

Kuo and Kawachi (2023) examine the role of structural socioeconomic conditions in shaping deaths of despair, focusing on county-level income inequality and intergenerational social mobility in the United States. Using data from 2000–2019, they find that both higher income inequality and lower social mobility are independently associated with increased mortality from drug overdoses, suicide, and alcohol-related liver disease across White, Black, and Hispanic populations. Importantly, their analysis shows that these factors interact synergistically: counties characterized by both high inequality and limited economic mobility experience disproportionately higher rates of deaths of despair than would be expected from either factor alone. The authors interpret these findings as evidence that unequal economic environments combined with limited opportunities for upward mobility generate psychosocial stress—such as feelings of relative deprivation and reduced hope—which may lead to harmful coping behaviors and increased mortality. Overall, the study provides strong support for the view that deaths of despair are linked to broader structural and socioeconomic conditions, highlighting the importance of addressing inequality and opportunity constraints in understanding and responding to the crisis.

Beseran et al. (2022) conduct a scoping review of the social determinants of “deaths of despair” (DoD) in the United States. Synthesizing 21 empirical studies, they document a sustained increase in DoD since the early 2000s and emphasize its strong association with widening social and economic inequalities (Beseran et al., 2022). The review identifies economic conditions as a primary driver. Economic insecurity, income decline, and negative shocks such as deindustrialization and trade liberalization are consistently linked to higher DoD mortality (Knapp et al., 2019; Pierce & Schott, 2016; as summarized in Beseran et al., 2022). Financial losses and deteriorating housing wealth also contribute to deaths of despair, as asset loss increases risk while gains in housing wealth reduce drug-related mortality (Fishman & Gutin, 2021; Jou et al., 2020; as summarized in Beseran et al., 2022). Redistributive policies such as minimum wage increases reduce suicide risk among low-educated individuals (Dow et al., 2020; as summarized in Beseran et al., 2022).

Labor market conditions and occupational characteristics constitute another key pathway. High-risk occupations, particularly blue-collar and precarious jobs, exhibit elevated DoD mortality due to job insecurity, hazardous working conditions, and non-standard employment arrangements (Gutin & Hummer, 2020; Hawkins et al., 2021; as summarized in Beseran et al., 2022). Unemployment and labor force exit are especially strong predictors, although elevated risks are also observed in some high-status professions (e.g., healthcare), partly linked to workplace stress and access to substances.

Educational attainment is a central axis of inequality. Lower education levels are consistently associated with higher DoD mortality, particularly for opioid-related deaths, while higher education appears

protective (Case & Deaton, 2015; Geronimus et al., 2019; as summarized in Beseran et al., 2022). Part of this relationship operates through economic mechanisms, as financial loss mediates the link between low education and despair-related behaviors (Fishman & Gutin, 2021; as summarized in Beseran et al., 2022).

Geographic disparities further shape DoD patterns. Rural and non-metropolitan areas, especially in regions such as Appalachia and the U.S. South, experience disproportionately high mortality, reflecting weaker labor markets, reduced healthcare access, and greater social isolation (Monnat, 2020; Stein et al., 2017; as summarized in Beseran et al., 2022). Finally, the role of race and ethnicity remains contested. While early work emphasized non-Hispanic Whites (Case & Deaton, 2015), more recent studies show increasing despair-related behaviors across Black and Hispanic populations as well, suggesting that DoD reflects broader structural inequalities rather than a single-group phenomenon (Gaydos et al., 2019; as summarized in Beseran et al., 2022).

In terms of structural factors through policy changes, Bradford and Bradford (2020) provide causal evidence linking housing instability proxied by eviction rates to deaths of despair, focusing on drug- and alcohol-related mortality in the United States. Using county-level panel data from 2004–2016, they merge eviction rates with cause-specific mortality records and estimate fixed-effects models with instrumental variables to address endogeneity. The authors find that higher eviction rates are consistently associated with higher substance-related mortality, particularly for opioids (including heroin and synthetic opioids), benzodiazepines, stimulants, and alcohol poisoning. Importantly, the results are concentrated in urban counties, with little evidence of similar effects in suburban or rural areas.

Despite the growing literature on the determinants of deaths of despair, an important gap remains in understanding the role of policy interventions and institutional changes. Much of the existing work has focused on structural socioeconomic factors such as labor market conditions, education, economic conditions, occupational features, and inequality, or on supply-side dynamics in drug markets. While these studies provide valuable insights into underlying drivers, relatively few directly evaluate how discrete policy changes, particularly in housing, labor market regulation, or social welfare programs, affect deaths of despair. This gap highlights the need for further research that explicitly links policy variation to deaths of despair, enabling clearer identification of actionable mechanisms and more effective policy design.

2.3 Labor Market Institutions and Mental Health

A growing body of research examines the interaction between labor market conditions and mental health. Jolivet and Postel-Vinay (2025) develop a structural life-cycle model that captures the two-way relationship between labor market outcomes and mental health. Using British panel data matched with occupational measures of job stress, their model shows that job loss and exposure to stressful work environments have lasting negative effects on both employment trajectories and psychological well-being. Their findings support the hypothesis that labor market deregulation by increasing job stress and insecurity can contribute to deteriorating mental health, which in turn affects labor supply and long-term career outcomes. This feedback loop reinforces the potential for cumulative disadvantage and despair.

Complementing structural models that link job characteristics to mental health, Gleit et al. (2024) reassess the “deaths of despair” hypothesis by comparing the relative importance of subjective distress and structural factors such as employment status and social integration in predicting mortality from drugs, alcohol, and suicide. Using longitudinal U.S. data with cause-specific mortality follow-up, they find little evidence that psychological or economic distress is uniquely associated with deaths of despair, as these factors predict mortality across a wide range of causes with similar magnitude. In contrast, structural conditions exhibit much stronger and more distinctive associations. Individuals detached

from the labor force or with weak social ties face substantially higher risks of death from drugs, alcohol, and suicide, with effects that exceed those observed for many other causes of mortality. Moreover, the relationship between distress and deaths of despair appears to be largely driven by confounding with structural disadvantage and pre-existing health conditions. Overall, the authors conclude that these deaths are better understood as consequences of social and economic disconnection rather than purely subjective psychological despair, emphasizing the importance of structural determinants in shaping self-destructive behaviors.

Building on this emphasis on structural determinants, Scheiring et al. (2020) further conceptualize deaths of despair as a consequence of deindustrialization, framing industrial decline as a “fundamental cause” of ill health. Using qualitative evidence from post-socialist Hungary, the authors show that deindustrialization generates persistent psychosocial stress and erodes coping resources through multiple interrelated mechanisms, including economic deprivation, precarious work conditions, the disruption of work and community identities, rising social inequality, and the collapse of local services and social networks. These processes jointly undermine individuals’ sense of stability, purpose, and social belonging, gradually producing feelings of powerlessness and hopelessness that can manifest in harmful health behaviors such as substance abuse. Importantly, the authors argue that despair should be understood as an outcome of prolonged structural dislocation rather than its primary cause, and that behavioral or lifestyle-based explanations are insufficient to account for observed mortality patterns. Overall, their findings reinforce the view that the roots of deaths of despair lie in broader economic transformations that reshape social environments and generate enduring health inequalities.

Extending this focus on structural determinants, Hawkins et al. (2021) highlight the importance of occupation-specific conditions in shaping deaths of despair. Using linked death certificate and labor market data from Massachusetts, the authors show that workers in occupations characterized by high injury and illness rates, greater job insecurity, and increasing reliance on non-standard work arrangements face substantially higher mortality risks, particularly from opioid-related causes. These patterns suggest multiple pathways through which workplace environments influence health, including injury-related opioid prescription, exposure to job loss and financial instability, and heightened psychosocial stress. Although some of these associations are attenuated after accounting for demographic factors such as education, the findings underscore the role of workplace structures in mediating health risks. Overall, the study demonstrates that beyond broad macroeconomic changes, the conditions of work itself such as hazardous, insecure, and precarious employment constitute a key mechanism linking economic transformations to deaths of despair.

Building on efforts to reconcile competing explanations, Bjorklund (2023) emphasizes the interaction between economic, social, and environmental factors in shaping deaths of despair. Using county-level data, the study finds that economic precarity—measured through indicators such as unemployment, poverty, and housing strain—is a strong predictor of mortality, while opioid availability and the concentration of the white working-class also exert independent effects. Importantly, the analysis highlights that these factors do not operate in isolation: opioid availability amplifies the adverse effects of economic insecurity, and the highest mortality rates occur in contexts where precarity, high opioid exposure, and large white working-class populations intersect. This interactive framework challenges single-cause explanations, suggesting instead that deaths of despair arise from a multifaceted process in which structural economic conditions, drug supply, and community-specific dynamics jointly reinforce one another. The embracement of both demand-side and supply-side of drug environments in this study is an appropriate direction from the aforementioned debate between the “death of despair” hypothesis and drug availability as a main driver of deaths of despair.

Taken together, these contributions point to several interconnected mechanisms through which labor market institutions shape mental health and deaths of despair. First, increased job insecurity,

unemployment risk, and exposure to high-stress or hazardous working conditions, often associated with deregulation and labor market flexibilization, generate persistent psychological strain that directly deteriorates mental well-being and indirectly reduces labor supply through feedback effects (Hawkins et al., 2021; Jolivet & Postel-Vinay, 2025). Second, structural dislocation processes such as deindustrialization weaken stable employment opportunities while eroding social identities and community cohesion, leading to chronic stress, loss of purpose, and diminished coping capacity (Scheiring et al., 2020). Third, labor market detachment and precarious employment are closely linked to social isolation and reduced social integration, which amplify both mental health risks and vulnerability to self-destructive behaviors (Glei et al., 2024). Finally, these economic and psychosocial stressors interact with contextual factors—including opioid availability and local socioeconomic composition—to magnify mortality risks, particularly in environments where economic precarity and access to harmful substances coexist (Bjorklund, 2023). Overall, the literature suggests that labor market institutions affect mental health and deaths of despair through a combination of material insecurity, psychosocial stress, social fragmentation, and interaction with health-risk environments, rather than through any single isolated pathway.

3 Conceptual Framework

This section develops the theoretical foundation for analyzing how employment protection deregulation may contribute to rising deaths of despair. It integrates insights from labor economics, public health, and sociology to trace the causal pathways linking institutional labor market changes to adverse health outcomes.

3.1 Employment Protection Legislation Deregulation

Employment protection legislation (EPL) refers to laws and regulations governing hiring and firing, job security, and dismissal procedures. Deregulation involves relaxing these protections, thereby increasing flexibility in labor markets.

Conceptually, EPL deregulation primarily affects the degree of job security faced by workers. Reductions in dismissal costs and procedural constraints increase exposure to job loss, income instability, and precarious employment. As emphasized in the literature, these changes can alter workers’ economic security, attachment to employment, and exposure to labor market risk, which constitute key inputs into the mechanisms outlined in Section 3.3.

Empirically, this paper focuses on the job-security dimension of EPL using the OECD Employment Protection Indicators. We use version 2 of the dataset (1998–2019) to ensure consistency with WHO mortality data under ICD-10 classification. The indicator captures the strictness of regulation on individual and collective dismissals and serves as the basis for identifying major deregulation episodes.

3.2 Deaths of Despair

The term “deaths of despair,” popularized by Case and Deaton (2020), refers to mortality from suicide, drug use disorders, and alcohol-related diseases. These causes of death have been linked to a range of underlying economic and social factors, including labor market conditions, income instability, and broader structural disadvantage.

While a large body of work emphasizes economic insecurity, loss of stable employment, and psychosocial stress as key drivers, other research highlights the importance of supply-side factors such as drug availability and institutional context. Moreover, the three components of deaths of despair exhibit

distinct patterns and may operate through partially different mechanisms. This heterogeneity motivates analyzing them separately in the empirical framework.

Key channels identified in the literature include economic insecurity, labor market detachment, and psychosocial stress, which jointly shape individuals' health behaviors and mortality risks.

3.3 Mechanisms Linking Deregulation to Despair

Building on Sections 2.2 and 2.3, the relationship between employment protection deregulation and deaths of despair operates through three core mechanisms that combine economic, psychosocial, and structural channels.

1. Economic Insecurity and Material Strain Deregulation increases job instability, unemployment risk, and income volatility. The literature identifies these forms of economic insecurity as fundamental drivers of adverse health outcomes, operating through financial strain, reduced future prospects, and heightened uncertainty. These conditions contribute to deteriorating mental and physical health and increase the likelihood of risky or self-destructive behaviors.

2. Labor Market Disconnection and Psychosocial Stress Weaker job protection increases precarious employment and detachment from stable work, eroding social integration and undermining identity and purpose. These processes generate chronic stress and increase the likelihood of harmful coping behaviors.

3. Interaction with Structural and Institutional Contexts The effects of deregulation depend on broader structural conditions. Economic shocks interact with health systems, inequality, and exposure to harmful environments, shaping how they translate into mortality.

4 Data

This section describes the data sources, variable construction, and sample used in the empirical analysis. We combine cross-country information on employment protection legislation, mortality outcomes, and institutional characteristics for a panel of OECD economies over 1998–2019.

4.1 OECD Indicators of Employment Protection

We measure employment protection using the OECD Employment Protection Indicators, which capture the strictness of regulations governing dismissals and the use of temporary contracts on a scale from 0 (least strict) to 6 (most strict). Consistent with the conceptual framework, we focus on the dimension of job security, specifically regulations on individual and collective dismissals.

Some key components of individual dismissals constructing the OECD indicators of employment protection are procedural requirement like advance notice periods, financial cost like severance pay obligations, and legal enforceability like access to legal recourse for employees. Key components of collective dismissals in this OECD indicators of employment protection are thresholds for triggering collective dismissal rules, consultation obligations, notification to public authorities, and additional procedural delays.

To ensure consistency with mortality data classified under ICD-10, we use version 2 of the OECD indicators, which provides annual data for the period 1998–2019. Our analysis exploits large discrete changes in these indicators to identify policy reforms. Following Ciminelli et al. (2022), major EPL reforms are defined as changes in the indicator that fall within the top or bottom 5th percentile of the

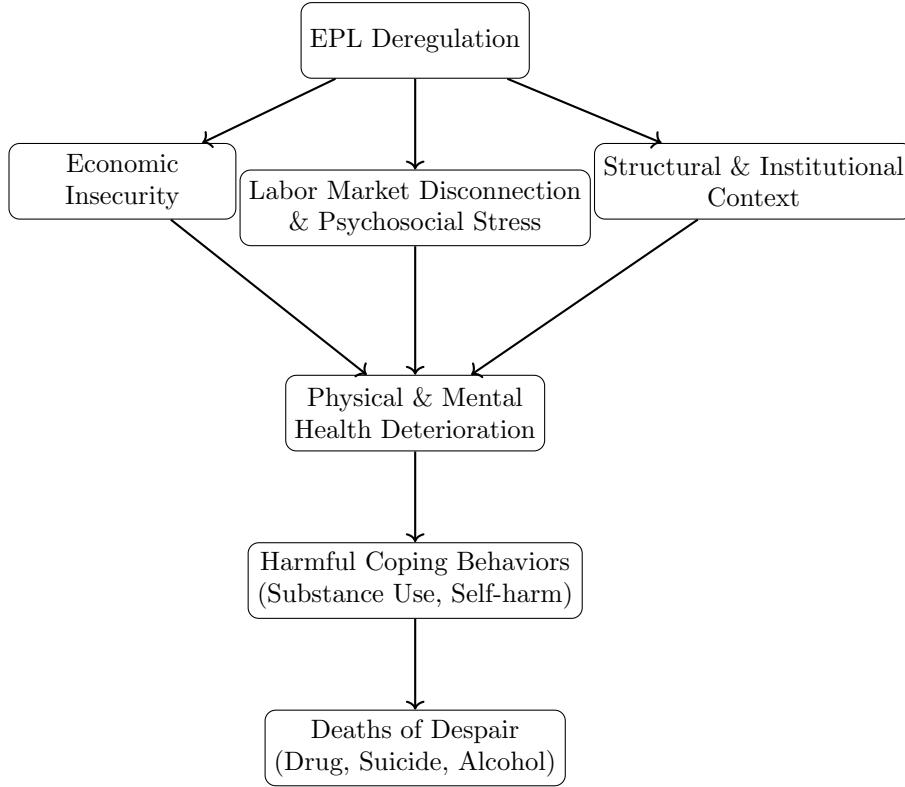


Figure 1: **Conceptual pathways linking employment protection deregulation to deaths of despair.**

distribution of annual changes. We construct an indicator variable that takes the value 1 for major deregulation reforms, -1 for major tightening reforms, and 0 otherwise. This approach isolates substantial policy shifts that are plausibly exogenous to short-term fluctuations in mortality.

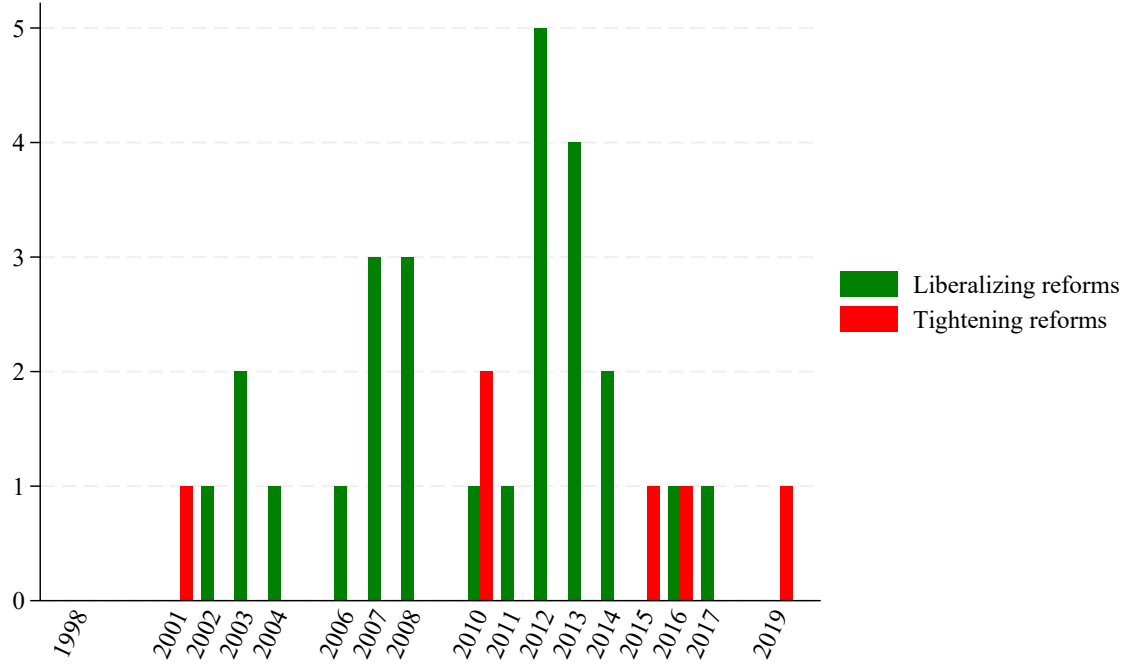
Figure 2 illustrates the timing and distribution of major EPL reforms across the sample period and countries. The top panel shows the aggregate number of liberalizing and tightening reforms by year. Reforms are unevenly distributed over time, with clear clustering in the mid-2000s and early 2010s. In particular, liberalizing reforms are most frequent between 2006 and 2013, with a pronounced peak around 2012, while tightening reforms are relatively infrequent and occur sporadically throughout the sample period. This pattern indicates that EPL changes occur in discrete reform episodes rather than as gradual adjustments.

The bottom panel displays the distribution of reforms by country, revealing substantial cross-country heterogeneity. Several countries, such as Portugal, Italy, and Spain, experience multiple liberalizing reforms concentrated within relatively short periods, whereas others exhibit few or no major reforms over the sample. Tightening reforms are less common and limited to a smaller set of countries, including Australia, Belgium, the Netherlands, and New Zealand. The variation in the timing and intensity of reforms across countries provides the key source of identifying variation for estimating the dynamic effects of EPL deregulation.

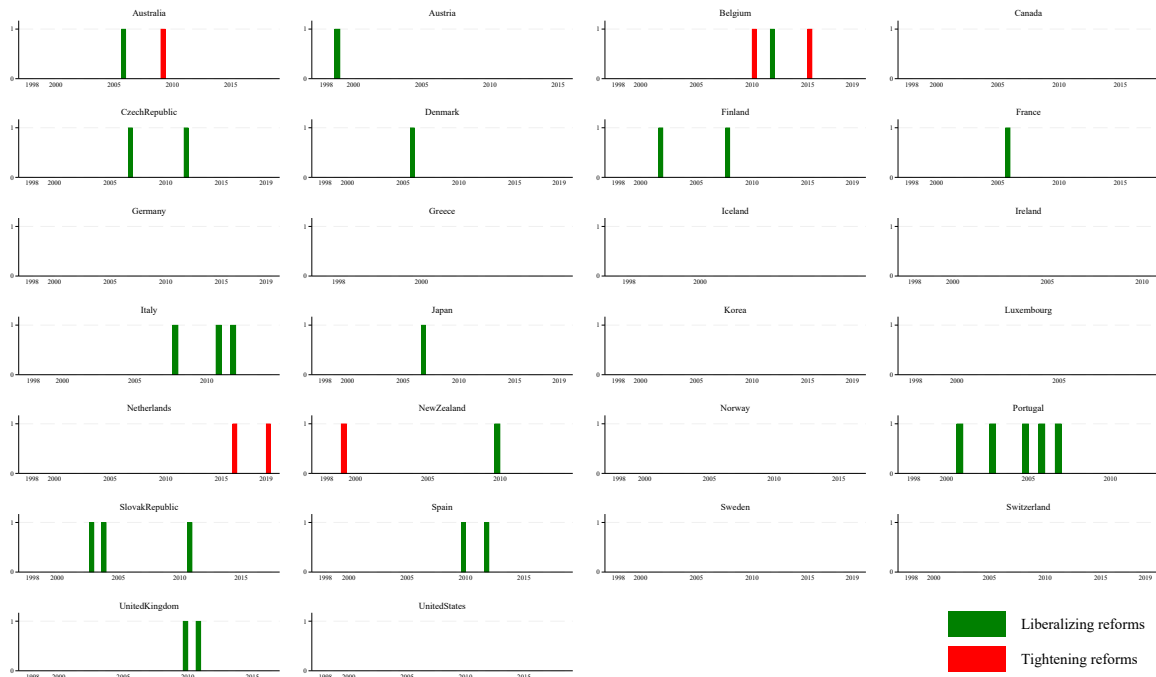
4.2 WHO Mortality Database

Mortality data are drawn from the World Health Organization (WHO) Mortality Database, providing age-standardized death rates per 100,000 population. Age standardization ensures comparability across countries with different demographic structures.

We focus on mortality causes associated with deaths of despair, using ICD-10 classifications. Specifi-



(a) Distribution of reforms over time



(b) Reforms by country

Figure 2: **Major employment protection reforms.** The top panel shows the aggregate distribution of reforms over time, while the bottom panel presents the corresponding country-level patterns.

cally, the main outcomes include deaths due to drug use disorders, alcohol use disorders, and self-inflicted injuries (suicide). Additional outcomes include violence and inflammatory heart diseases, which are plausibly linked to substance use and prolonged stress.

To ensure consistency and avoid measurement discontinuities, we restrict the sample to ICD-10 data. Many countries experienced substantial breaks in mortality series during the transition from ICD-9 to ICD-10, making pre-transition data less comparable. Since most OECD countries adopted ICD-10 around 1998, this restriction yields a balanced and reliable panel covering 26 countries over 1998–2019. Figure 1 shows the number of countries and areas with data in the WHO Mortality Database by ICD code and year, as of April 2022.

The main dependent variables in results include deaths attributed to:

- Drug use disorders (ICD-10: F11-F16, F18-F19, X41-X42, X44),
- Alcohol use disorders (ICD-10: F10, X45),
- Self-inflicted injuries (ICD-10: X60-X84, Y870),
- Violence (ICD-10: X85-Y09, Y871),
- Inflammatory heart diseases (ICD-10: I30-I33, I38, I40, I42).
- Cirrhosis of liver (ICD-10: K70, K74)

The sample includes 26 OECD countries from 1998 to 2019.

4.3 OECD Country-specific Characteristics

To examine heterogeneous effects, we incorporate country-specific characteristics capturing key institutional and structural conditions. These variables enter the empirical specification as interaction terms with the EPL reform indicator and are normalized to represent differences between the 25th and 75th percentiles.

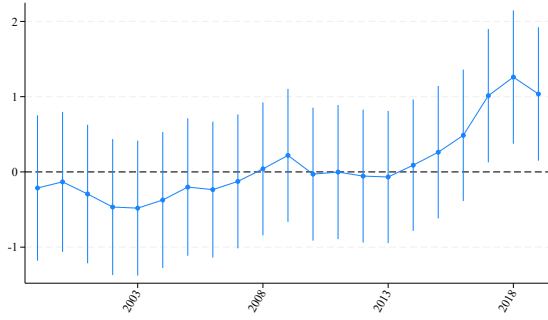
We consider three broad categories of moderators. First, income support is measured using unemployment benefit replacement rates and government social transfer expenditures, which proxy the extent of economic insurance. Second, health system capacity is captured by avoidable mortality rates and public health insurance coverage, reflecting the effectiveness and accessibility of healthcare systems. Third, broader social conditions are proxied by poverty rates, which capture structural disadvantage.

Avoidable mortality can represent the effectiveness of public health and healthcare system, which includes both preventable and treatable deaths. Preventable deaths are defined as causes of death among people aged under 75 years that can be avoided through effective public health and prevention interventions (i.e. before the onset of disease/injury, to reduce incidence). Treatable deaths are defined as causes of death that can be avoided through timely and effective healthcare interventions, including secondary prevention and treatment (i.e. after the onset of disease, to reduce case fatality) (OECD, 2023).

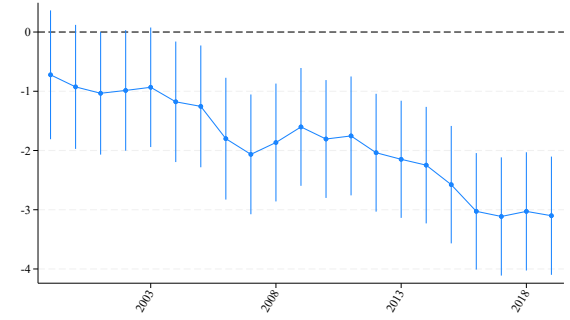
These measures allow us to assess whether stronger social protection and health institutions mitigate the impact of EPL deregulation on deaths of despair, consistent with the institutional buffering mechanism outlined in Section 3.

4.4 Stylized Facts in Mortality Rates of Deaths of Despair

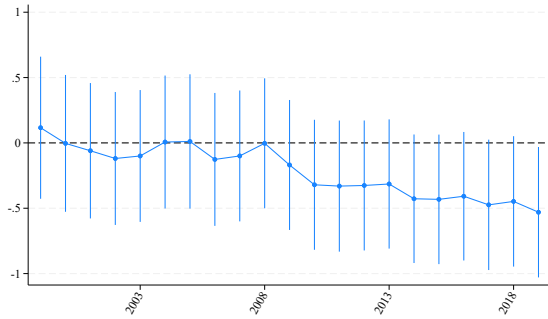
Figure 3 summarizes stylized facts on cause-specific mortality across OECD countries. It reports average changes in mortality rates by year, obtained from a specification with country and year fixed effects.



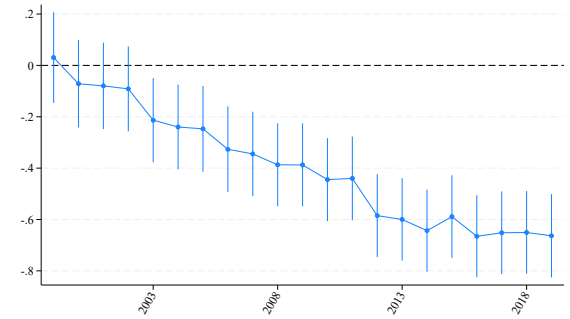
(a) Drug



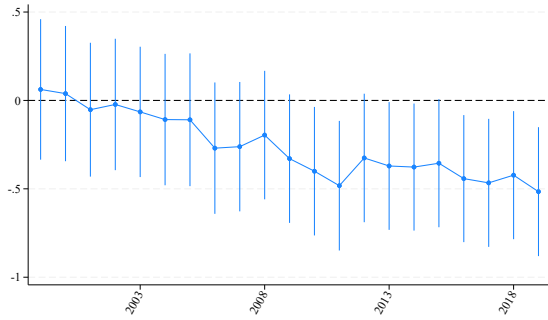
(b) Suicide



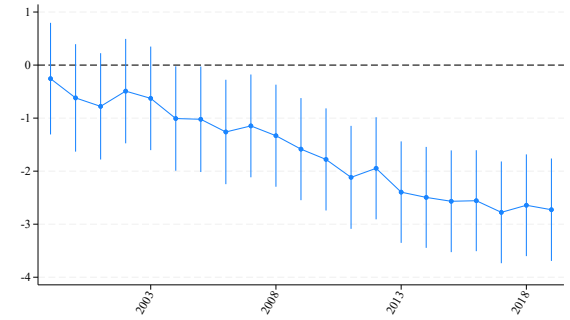
(c) Alcohol



(d) Violence



(e) Inflammatory Heart Disease

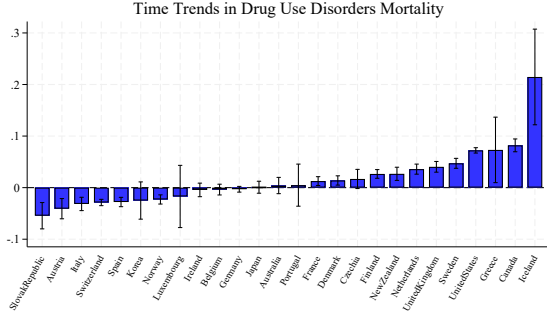


(f) Cirrhosis of Liver

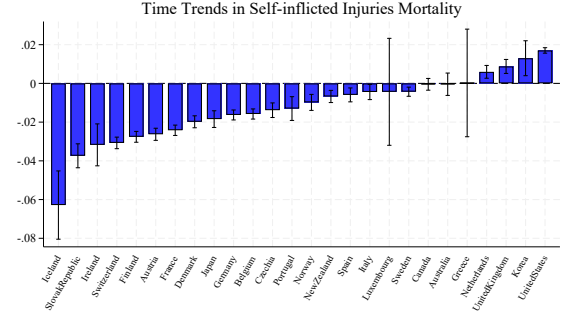
Figure 3: **Average changes in cause-specific mortality rates across 26 OECD countries.** Each panel reports the coefficients on the year fixed effects from the regression

$$Mortality_{i,t} = \alpha + \tau_t + \gamma_i + \varepsilon_{i,t},$$

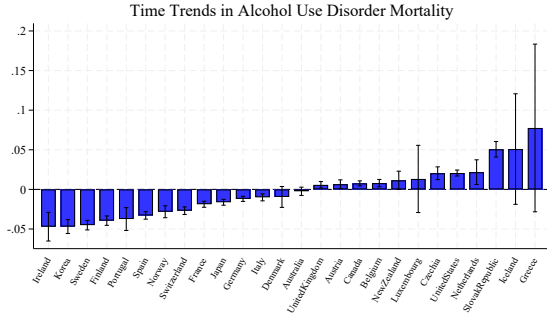
where i and t index country and year, respectively. $Mortality$ denotes the cause-specific mortality rates, α is a constant, τ_t are year fixed effects, and γ_i are country fixed effects. Vertical lines indicate 95% confidence intervals. The estimates can be interpreted as the average change in mortality rates relative to 1998, the omitted base year.



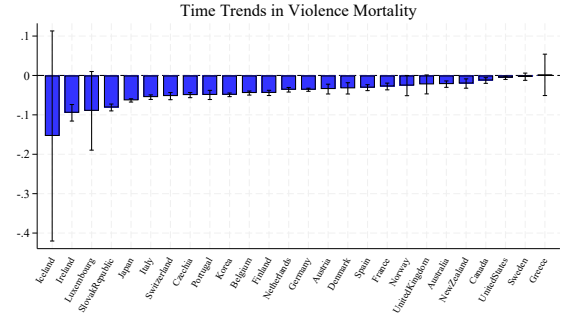
(a) $\ln(\text{Drug})$



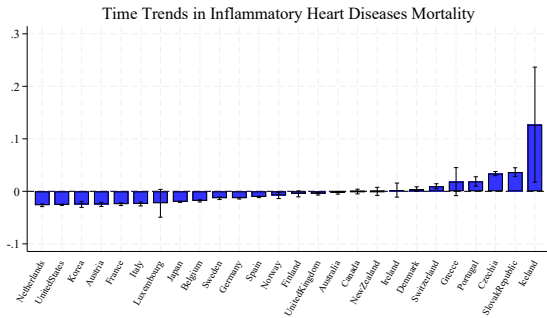
(b) $\ln(\text{Suicide})$



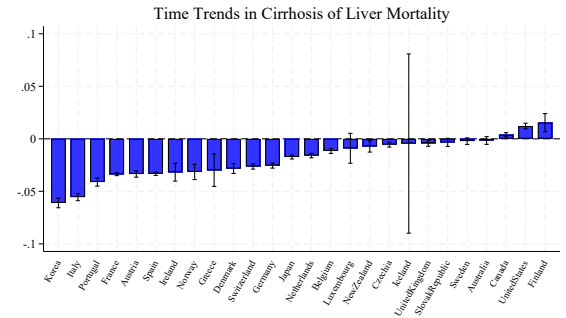
(c) $\ln(\text{Alcohol})$



(d) $\ln(\text{Violence})$



(e) $\ln(\text{Inflammatory Heart Disease})$



(f) $\ln(\text{Cirrhosis of Liver})$

Figure 4: **Country-specific time trends in log mortality rates by cause.** Each panel shows estimated linear trends in log age-adjusted mortality for the indicated cause of death of each country. Trends are obtained from the regression

$$Mortality_t^i = \alpha^i + \tau^i t + \varepsilon_t^i,$$

where the subscript t and superscript i denote year and country, respectively. $Mortality$ denotes the cause-specific mortality rates, α^i is a country-specific constant, τ^i is the country-specific linear time trend, and ε_t^i is an error term. Capped spikes indicate 90 percent confidence intervals. The estimates can be interpreted as the average annual change in a cause-specific mortality rate for each country over the sample period, which for most countries spans 1998–2019.

Several common patterns emerge. Only drug use disorder mortality exhibits a marked upward trend over the sample period, with increases especially pronounced after the mid-2000s. In contrast, alcohol use disorder and inflammatory heart disease mortality displays a gradual decline over time, while suicide, violence mortality and cirrhosis of liver mortality show statistically significant and persistent downward trends. These aggregate patterns highlight substantial heterogeneity across causes of death, motivating their separate analysis in the empirical framework.

Figure 4 illustrates country-specific time trends in log mortality rates for each cause. The estimates reveal considerable cross-country heterogeneity in both the magnitude and direction of trends. For drug use disorders, countries exhibit mixed trends with positive and negative changes although positive changes are more significant and outstanding. Suicide mortality trends are also mixed, with a few countries experiencing increases and most countries showing significant decreases. Mortality rates from cirrhosis of liver show significant negative trends in most countries with a very few with increasing trends. Similarly, alcohol use disorder mortality indicates mixed trends but with more significant negative trends and insignificant positive trends caused by very large standard deviations. In contrast, violence mortality displays predominantly negative trends across countries, consistent with the aggregate patterns. On the other hand, inflammatory heart disease mortality displays relatively modest trends in both positive and negative direction. This heterogeneity underscores the importance of exploiting both cross-country and time-series variation to identify the effects of EPL reforms.

5 Empirical Methodology

This section describes the empirical strategy used to estimate the dynamic effects of employment protection legislation (EPL) reforms on deaths of despair. To capture the timing and persistence of these effects while avoiding restrictive functional form assumptions, we employ the local projection method introduced by Jordà (2005).

5.1 Baseline Specification

For each horizon $k = 0, \dots, 5$, we estimate the following specification:

$$y_{i,t+k} - y_{i,t-1} = \alpha_i + \tau_t + \beta_k R_{i,t} + \theta \mathbf{X}_{i,t} + \sum_{h=1}^k \phi_h R_{i,t+h} + \sum_{l=1}^4 (\delta_l \Delta y_{i,t-l} + \gamma_l R_{i,t-l}) + \varepsilon_{i,t}. \quad (1)$$

The dependent variable, $y_{i,t+k}$, denotes the age-standardized mortality rate for a given cause of death in country i at horizon k . The specification models cumulative changes in mortality relative to the pre-reform period, allowing the coefficients β_k to be interpreted as impulse responses to EPL reforms.

The variable $R_{i,t}$ is an indicator of major EPL reform, capturing discrete policy changes identified from the OECD Employment Protection Indicators. Country fixed effects (α_i) control for time-invariant heterogeneity across countries, while year fixed effects (τ_t) capture global shocks affecting all countries simultaneously. The vector $\mathbf{X}_{i,t}$ includes standard macroeconomic controls.

To isolate the effect of reforms occurring at time t , the specification includes forward reform terms $\sum_{h=1}^k \phi_h R_{i,t+h}$, which account for additional reforms implemented within the response horizon. In addition, four lags of both the outcome variable and the reform indicator are included to mitigate concerns about reverse causality and dynamic adjustment.

5.2 Interpretation

The coefficients β_k trace the dynamic response of mortality to a major EPL reform over time. By estimating the model separately for each horizon k , we obtain impulse response functions (IRFs) that describe how mortality evolves in the years following a reform. Standard errors are clustered to account for serial correlation.

5.3 Heterogeneous Effects

To examine how institutional and structural conditions shape the impact of EPL reforms, we extend the baseline specification to include interaction terms:

$$y_{i,t+k} - y_{i,t-1} = \alpha_i + \tau_t + \beta_{1k}(R_{i,t} \times S_{i,t}) + \beta_{2k}R_{i,t} + \beta_{3k}S_{i,t} + \cdots + \varepsilon_{i,t}. \quad (2)$$

Here, $S_{i,t}$ represents country-specific characteristics, including measures of income support, health system capacity, and social conditions. These variables are normalized to reflect differences between countries at the 25th and 75th percentiles.

The interaction coefficient β_{1k} captures how the effect of EPL reforms varies across institutional environments. Specifically, it can be interpreted as the differential impact of a reform between countries with relatively stronger and weaker social and health protections.

5.4 Identification

The empirical strategy relies on the assumption that major EPL reforms are plausibly exogenous to short-term fluctuations in mortality. This assumption is supported by the fact that such reforms are typically driven by political processes, long-term labor market considerations, or external policy pressures rather than contemporaneous health shocks.

The identification therefore exploits variation in the timing of reforms across countries and over time, comparing the evolution of mortality before and after reform episodes relative to countries that do not reform. The inclusion of fixed effects, lagged outcomes, and forward reform controls further helps isolate the causal effect of EPL deregulation.

6 Results

6.1 Main Effects

Figure 5 reports the dynamic effects of major EPL deregulation on cause-specific mortality, estimated using the local projection framework. Two key patterns emerge.

First, EPL deregulation leads to a statistically and economically significant increase in mortality from drug use disorders. The effect is not immediate but builds gradually over time, becoming pronounced several years after reform. At the five-year horizon, drug-related mortality increases by approximately 35 percent relative to pre-reform levels, indicating substantial medium-run effects.

Second, suicide mortality also rises following deregulation, although the magnitude is smaller. The effect becomes statistically significant within a few years and reaches roughly 5 percent at the five-year horizon. In contrast, the estimated effect on alcohol-related mortality is not statistically distinguishable from zero throughout the post-reform period.

These results highlight important heterogeneity across components of deaths of despair, suggesting that labor market deregulation primarily affects outcomes linked to mental health and substance use rather than alcohol-related diseases more broadly.

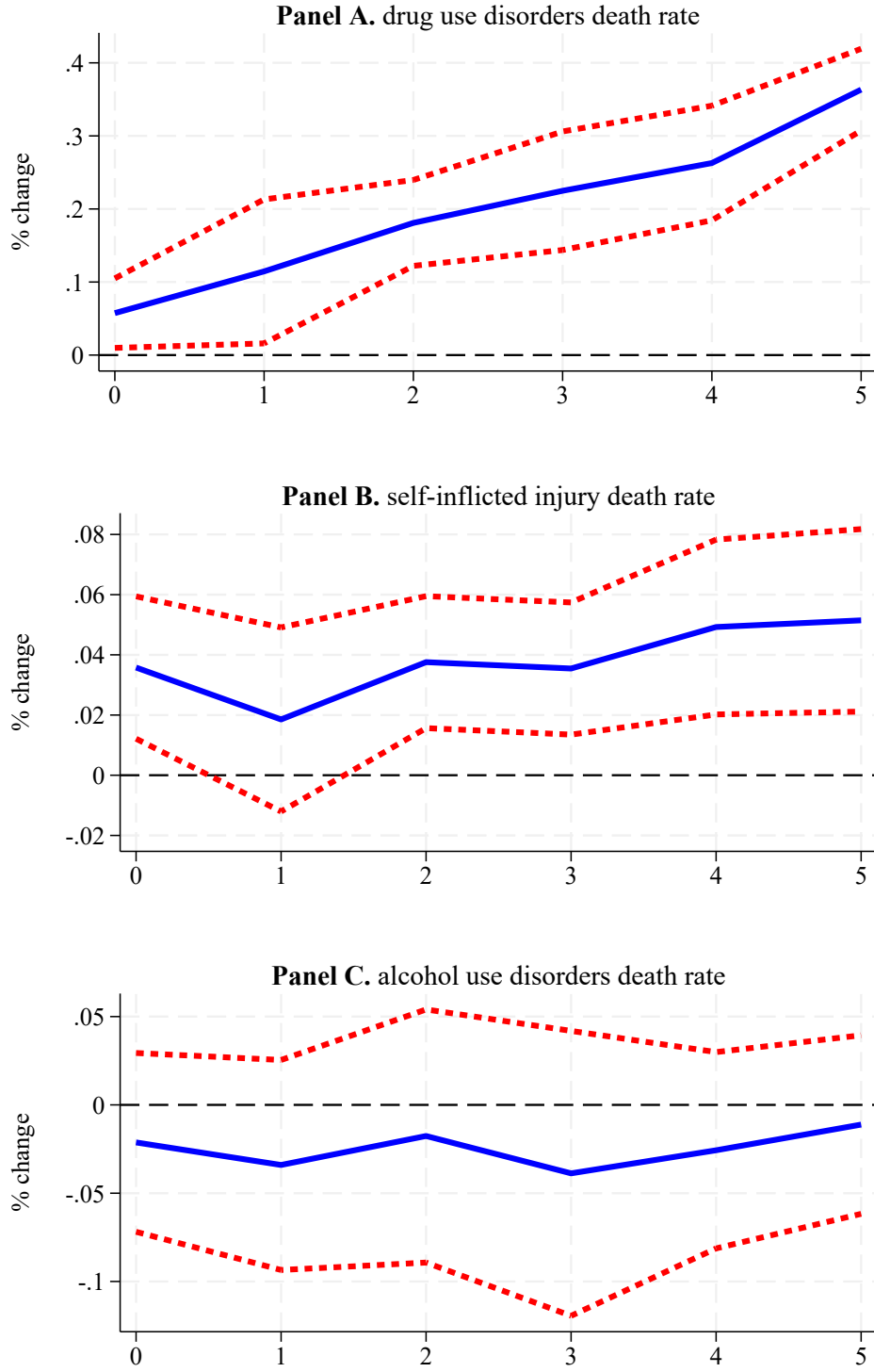


Figure 5: **Dynamic effects of major EPL deregulation on deaths of despair.** Each panel plots local-projection estimates of the post-reform response of the log mortality rate (per 100,000 population) for: (A) drug use disorders; (B) self-inflicted injury (suicide); and (C) alcohol use disorders. The horizontal axis reports years since reform (horizons $h = 0, \dots, 5$). Solid lines show point estimates; shaded bands are 95% confidence intervals. All specifications include country and year fixed effects. Standard errors are Driscoll–Kraay, robust to heteroskedasticity, autocorrelation, and cross-sectional dependence. Sample: 26 OECD countries, [1998-2019].

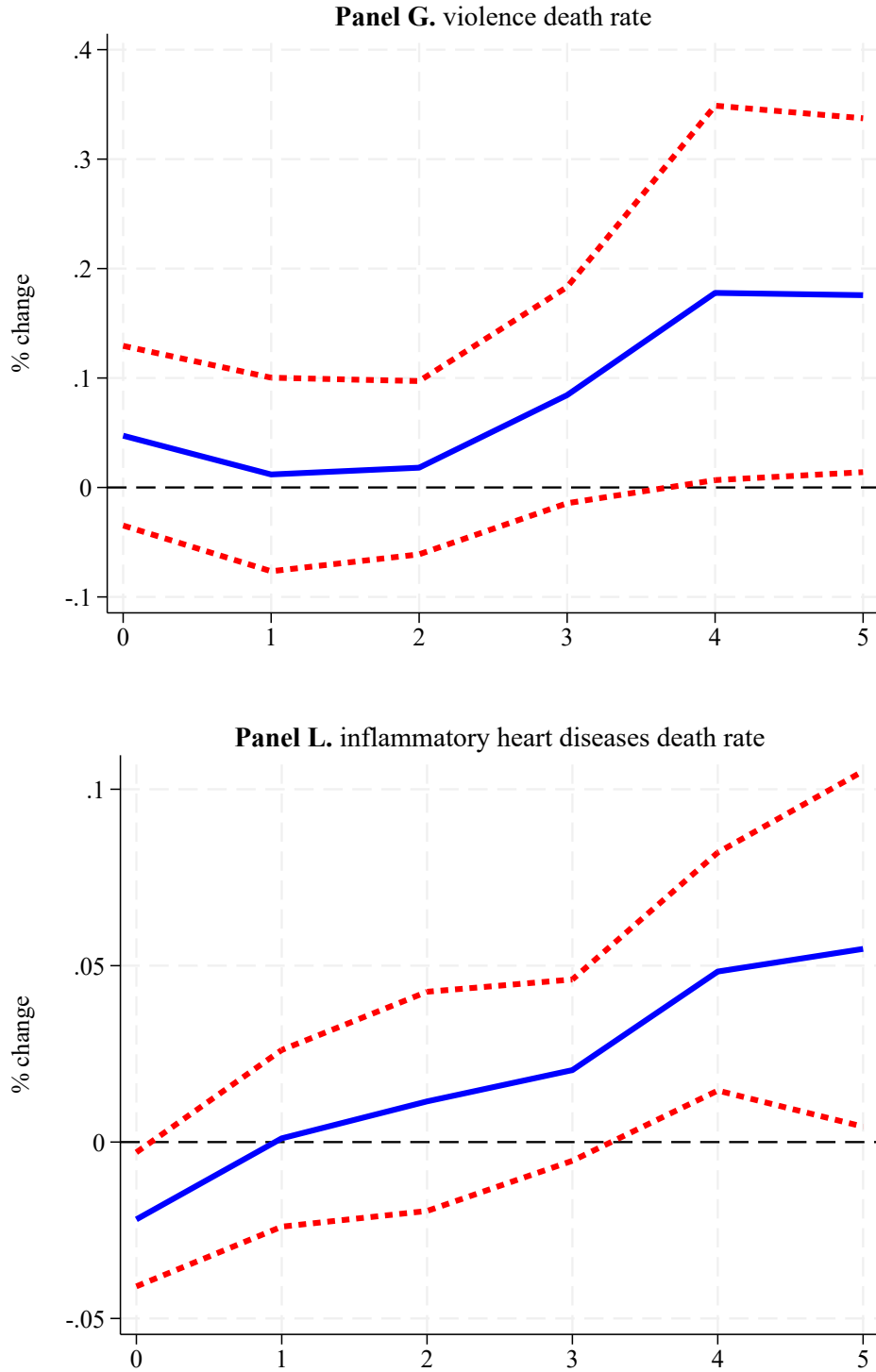


Figure 6: **Spillovers to violence and inflammatory heart diseases.** Panels show LP estimates of the post-reform response of the log mortality rate (per 100,000 population) for: (A) violence; and (B) inflammatory heart diseases. The horizontal axis reports years since reform (horizons $h = 0, \dots, 5$). Solid lines show point estimates; shaded bands are 95% confidence intervals. All specifications include country and year fixed effects. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, [1998-2019].

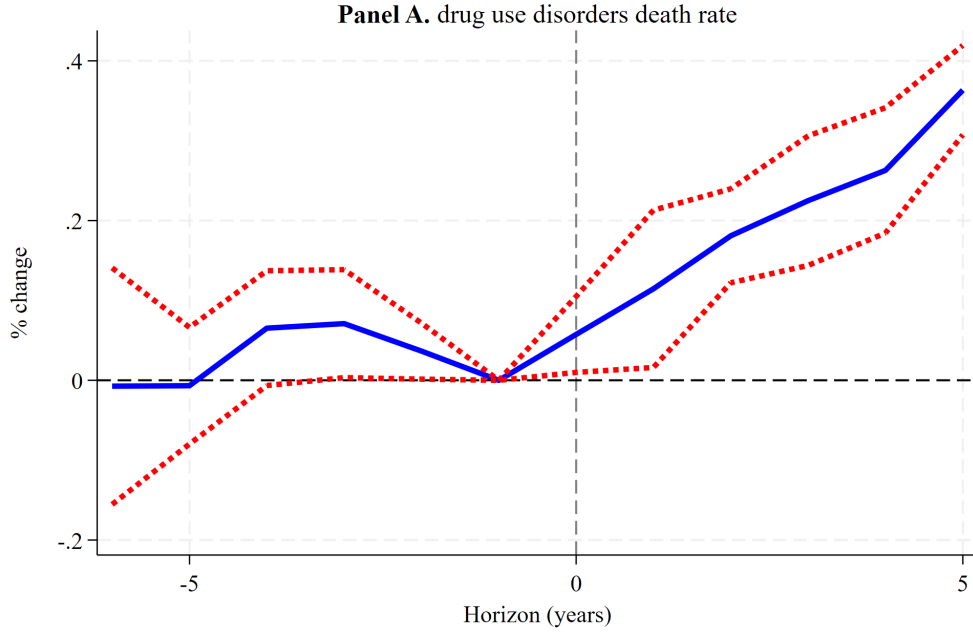


Figure 7: **Dynamic effects of major EPL deregulation on drug use disorder mortality with pre-trend.** Panels show LP estimates of both pre-trend and post-reform response of the log mortality rate (per 100,000 population) for: drug use disorder. The horizontal axis reports years before and after reform (horizons $h = -6, \dots, 0, \dots, 5$). Solid lines show point estimates; shaded bands are 95% confidence intervals. Standard errors are Driscoll–Kraay. Sample: 26 OECD countries, [1998-2019].

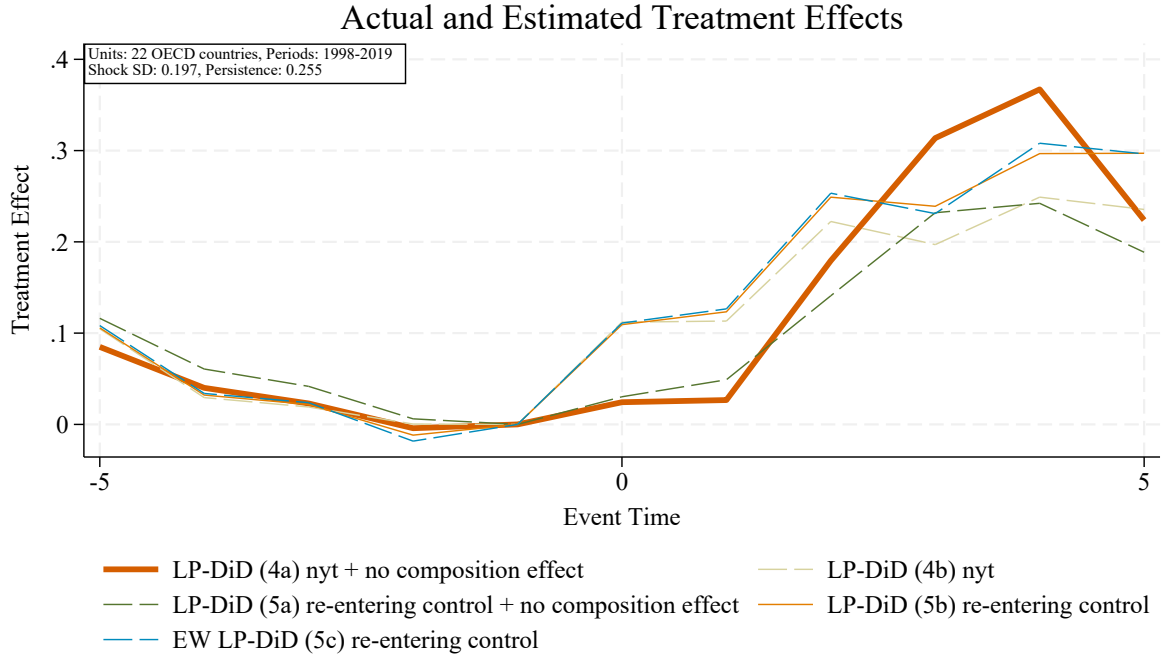


Figure 8: **Robustness check of the effects of major EPL deregulation on drug use disorder mortality with LP-DiD method.** This figure shows LP-DiD estimates of both pre-trend and post-reform response of the log mortality rate (per 100,000 population) for: drug use disorder. The horizontal axis reports years before and after reform (horizons $h = -5, \dots, 0, \dots, 5$). Solid lines show point estimates; shaded bands are 95% confidence intervals. Standard errors are Driscoll–Kraay. Sample: 22 OECD countries, [1998-2019].

6.2 Pre-trends and Identification

To assess the validity of the identification strategy, Figure 7 presents event-study estimates for drug use disorder mortality, including both pre- and post-reform periods. The pre-reform coefficients provide a direct test of the parallel trends assumption.

Overall, the pre-treatment estimates are close to zero and do not exhibit a systematic trend. While minor fluctuations are visible at longer leads, these are statistically indistinguishable from zero and do not display a consistent upward or downward pattern. This absence of differential pre-trends suggests that treated and untreated countries follow similar trajectories prior to reform.

In contrast, the post-reform coefficients display a clear and increasing pattern. The estimated effects rise steadily over time, consistent with a gradual accumulation of impacts following deregulation. Taken together, these findings support the interpretation that the observed post-reform increases in mortality are not driven by pre-existing trends but reflect the consequences of EPL reforms.

6.3 Robustness: LP-DiD Estimates

Figure 8 reports robustness checks based on the local projection difference-in-differences (LP-DiD) approach of Dube et al. (2025), estimated using only countries experiencing liberalizing reforms. This approach addresses potential concerns related to treatment heterogeneity, staggered adoption, and contamination from tightening reforms.

Across all specifications, the results exhibit a consistent dynamic pattern. Pre-reform coefficients remain close to zero, providing additional support for the absence of differential pre-trends. Following reform, the estimated treatment effects increase steadily, reaching magnitudes comparable to those obtained in the baseline specification.

Importantly, the results are robust across alternative implementations of the LP-DiD estimator, including specifications that account for re-entering control groups and alternative weighting schemes. Although some variation in magnitude arises at longer horizons, the qualitative pattern of increasing post-reform effects remains stable. These findings indicate that the baseline results are not driven by specification choice or sample composition.

6.4 Spillover Effects

We next examine spillovers to related causes of death. Figure 6 shows that EPL deregulation is associated with increases in mortality from violence and inflammatory heart diseases at longer horizons. These effects are not statistically significant in the short run but emerge several years after reform.

The timing and nature of these spillovers are consistent with indirect pathways linking economic insecurity to health outcomes. Increases in substance use and prolonged stress may contribute to both heightened exposure to violent environments and elevated cardiovascular risk.

6.5 Heterogeneous Effects by Institutional Context

7 Discussion

7.1 Plausible pathways of spillovers

There are three types of inflammatory heart diseases such as myocarditis, pericarditis, and endocarditis. Myocarditis is the inflammation of heart muscle and leads to weakening the heart’s ability to pump blood, arrhythmias, heart failure, or sudden death. Pericarditis is the inflammation of pericardium, a sac surrounding a heart. Endocarditis is the inflammation of the inner lining of heart and valves, which

Table 1: Dynamic effects of major EPL deregulation on mortality from drug use disorders

Controls for	0y	1y	2y	3y	4y	5y
Baseline	0.06** (0.0224)	0.11** (0.0462)	0.18*** (0.0274)	0.22*** (0.0375)	0.26*** (0.0360)	0.36*** (0.0254)
Crisis dummy	0.05** (0.0237)	0.11** (0.0412)	0.19*** (0.0276)	0.22*** (0.0330)	0.25*** (0.0455)	0.37*** (0.0283)
Lagged unemployment	0.05** (0.0209)	0.12** (0.0480)	0.19*** (0.0257)	0.24*** (0.0394)	0.27*** (0.0383)	0.37*** (0.0312)
Replacement rate of UB	0.06** (0.0223)	0.11** (0.0462)	0.18*** (0.0272)	0.22*** (0.0370)	0.26*** (0.0357)	0.36*** (0.0254)
Union density	0.05*** (0.0164)	0.11** (0.0474)	0.18*** (0.0271)	0.23*** (0.0350)	0.24*** (0.0545)	0.34*** (0.0431)
Manufacturing share of GDP	0.06** (0.0217)	0.11** (0.0470)	0.17*** (0.0251)	0.20*** (0.0369)	0.23*** (0.0321)	0.30*** (0.0313)
Social transfers expenditure	0.06** (0.0218)	0.11** (0.0443)	0.18*** (0.0261)	0.22*** (0.0343)	0.27*** (0.0422)	0.38*** (0.0445)

Notes: Outcome is the log mortality rate from drug use disorders (per 100,000 population). Entries report dynamic semi-elasticities at horizons $h = 0$ to 5 years after major EPL deregulation; standard errors in parentheses. Stars denote significance: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 2: Dynamic effects of EPL deregulation on mortality from drug use disorders: interaction terms

Interaction term with	0y	1y	2y	3y	4y	5y
Avoidable death rate	0.14** (0.0544)	0.22 (0.1279)	0.34** (0.1372)	0.50** (0.1699)	0.40** (0.1785)	0.40** (0.1731)
Social transfers expenditure	0.05 (0.0843)	0.14 (0.1615)	0.05 (0.2287)	0.17 (0.2829)	0.35 (0.3316)	0.49 (0.3240)
Public health insurance coverage	0.01*** (0.0036)	0.01* (0.0082)	0.04** (0.0156)	0.08** (0.0324)	0.11* (0.0573)	0.17* (0.0836)
Replacement rate of UB (60 months)	0.09 (0.0957)	0.22 (0.1887)	0.02 (0.2097)	0.05 (0.2237)	-0.06 (0.2350)	0.06 (0.2046)
Poverty rate	-0.19* (0.0947)	-0.47** (0.1934)	-0.54*** (0.1593)	-0.62** (0.2051)	-0.51** (0.1812)	-0.19 (0.1384)

Notes: Outcome is the **log mortality rate from drug use disorders** (per 100,000 population). Entries are interaction effects (P75–P25 style) at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

can destroy heart valves, cause emboli, and lead to severe complications. All three of them can be caused by the toxicity of alcohol and drug. For example, heavy chronic alcohol consumption is strongly associated with toxic myocarditis (Piano & Phillips, 2023). Drugs such as cocaine and amphetamines have a strong link to myocarditis due to catecholamine surge, vasospasm, and direct toxicity (Virmani et al., 2001). Opioid and IV drug use are the leading cause of infective endocarditis in young adults, and Schmidt et al. (2022) report a sharp rise in endocarditis cases among opioid users.

The relationship between violence mortality and drug or alcohol use disorders is documented in public health and criminology research. In terms of direct behavioral link, alcohol impairs judgment and lowers inhibitions, increasing the likelihood of violent behavior such as homicide and assault. Certain drugs like cocaine and methamphetamine can cause paranoia, impulsivity, and aggression, leading to violent incidents. In addition, illicit drug trade involves organized crime, territorial disputes, and enforcement violence, which all can contribute to violence mortality.

Epidemiological studies consistently demonstrate a strong association between substance use disorders and violent mortality. Alcohol use disorder significantly elevates the risk of homicide, with meta-analyses

indicating a three- to four-fold increase compared to individuals without alcohol dependence (Foran & O’Leary, 2008; Kuhns et al., 2014). Similarly, drug use disorders, particularly involving stimulants and opioids, are linked to heightened violence-related deaths, largely due to pharmacological effects and systemic violence in illicit drug markets (Darke, 2010; Kuhns et al., 2014). Toxicology reports reveal that approximately 40–50% of homicide victims and perpetrators have alcohol present at the time of the incident (World Health Organization, 2018). Cocaine and opioids are frequently detected among homicide victims, and opioid-related endocarditis and overdose crises have coincided with increasing violence in several regions (Darke, 2010; Schmidt et al., 2022). These findings underscore the dual role of substance use disorders as both behavioral risk factors and structural drivers of violent mortality.

7.2 Interpretation of Mechanisms

The results provide evidence that employment protection deregulation has meaningful downstream effects on mortality, particularly for drug use disorders and suicide. These findings are consistent with mechanisms operating through economic insecurity, labor market disconnection, and psychosocial stress.

Deregulation increases exposure to job loss, income volatility, and precarious employment, which can generate sustained economic and psychological strain. Over time, these pressures may lead to deteriorating mental health and increased reliance on maladaptive coping behaviors, including substance use and self-harm. The gradual emergence of effects observed in the data supports this interpretation, as the consequences of economic insecurity are likely to accumulate rather than materialize immediately.

7.3 Alternative Explanations

An important alternative explanation emphasizes supply-side factors, particularly changes in drug availability and potency. While such factors likely contribute to observed mortality trends, the cross-country design of this study suggests that labor market institutions play an independent role by shaping vulnerability to these risks.

In particular, the fact that the effects of EPL deregulation vary systematically with institutional characteristics—such as social transfers and health system capacity—indicates that the observed mortality increases cannot be fully explained by common supply shocks. Instead, labor market conditions appear to influence both exposure and resilience to harmful environments.

7.4 Institutional Buffering

The heterogeneity results highlight the importance of institutional context. Countries with stronger income support systems and more effective health care provision experience significantly smaller increases in mortality following deregulation.

These findings suggest that social protection and health systems act as buffers that mitigate the translation of economic shocks into adverse health outcomes. In this sense, labor market institutions should be viewed as part of a broader policy environment that jointly determines population health.

7.5 Policy Implications

The results have important policy implications. Reforms aimed at increasing labor market flexibility may generate unintended public health costs, particularly in settings with limited social protection. Policymakers considering such reforms should account for these externalities and consider complementary policies that strengthen income insurance and access to healthcare.

More broadly, the findings underscore the need to evaluate labor market policies not only in terms of employment or productivity outcomes, but also in terms of their effects on health and well-being.

7.6 Limitations

Several limitations should be noted. First, the analysis relies on country-level data, which does not allow for direct identification of individual-level mechanisms. Second, although the empirical design mitigates concerns about reverse causality, unobserved factors may still influence both reforms and mortality. Third, the role of drug supply dynamics is not directly measured and may interact with the mechanisms identified in this study.

7.7 Conclusion of Findings

Overall, the evidence supports a framework in which labor market institutions shape the relationship between economic conditions and mortality. The results are most consistent with an interpretation in which structural economic changes interact with institutional environments to generate deaths of despair, rather than a single dominant causal factor.

8 Conclusion

This paper shows that major employment-protection deregulation in advanced economies leads to a clear rise in deaths of despair—particularly drug-related and suicide mortality—within five years of reform. Using OECD EPL indicators and WHO mortality data for 26 countries from 1998–2019, the analysis finds that drug deaths increase by roughly 35 percent and suicide mortality by about 5 percent, with no significant effect on alcohol-related deaths. Spillovers into violence and inflammatory heart-disease mortality further underscore the broader health consequences of weakening dismissal protections.

The effects, however, are not uniform. Countries with stronger unemployment benefits, larger social-transfer systems, more comprehensive public-health-insurance coverage, and lower avoidable mortality experience substantially smaller increases in deaths of despair. Weaker safety nets and less effective health systems amplify the impact, supporting an institutional-buffering mechanism in which protections outside the labor market shape how economic insecurity translates into mortality risk.

These findings highlight that EPL reforms carry significant public-health implications that standard labor-market evaluations often overlook. Policymakers pursuing flexibility-enhancing reforms should pair them with stronger income supports, mental-health and addiction services, and robust health-system capacity to limit unintended harm. Future work should examine worker-level mechanisms, regional heterogeneity, and interactions with other structural policies such as housing protections. Overall, the results suggest that employment protection functions not only as a labor-market institution but as an important component of a country’s public-health infrastructure.

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10 Appendix

Table 3: Dynamic effects of EPL deregulation on mortality from self-inflicted injury: interaction terms

Interaction term with	0y	1y	2y	3y	4y	5y
Avoidable death rate	-0.0001 (0.0177)	-0.02 (0.0306)	-0.01 (0.0301)	-0.03 (0.0528)	-0.01 (0.0695)	-0.02 (0.0789)
Social transfers expenditure	0.04 (0.0286)	0.09** (0.0403)	0.06 (0.0506)	0.10 (0.0763)	0.16** (0.0662)	0.18** (0.0782)
Public health insurance coverage	-0.0018 (0.0011)	-0.0049** (0.0020)	-0.0031 (0.0030)	-0.0039 (0.0051)	-0.0038 (0.0129)	0.0008 (0.0184)
Replacement rate of UB (60 months)	0.02 (0.0187)	0.04 (0.0324)	0.02 (0.0397)	-0.01 (0.0467)	-0.01 (0.0594)	0.06 (0.0582)
Poverty rate	0.01 (0.0320)	0.00 (0.0570)	-0.04 (0.0675)	-0.01 (0.0691)	-0.04 (0.0790)	-0.01 (0.0557)

Notes: Outcome is the **log mortality rate from self-inflicted injury (suicide)** per 100,000 population. Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. If figures report percent changes, coefficients can be transformed as $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 4: Dynamic effects of EPL deregulation on mortality from alcohol use disorder: interaction terms

Interaction term with	0y	1y	2y	3y	4y	5y
Avoidable death rate	-0.02 (0.0379)	0.03 (0.0540)	0.03 (0.0743)	-0.01 (0.0955)	0.04 (0.1192)	0.11 (0.1170)
Social transfers expenditure	-0.18*** (0.0411)	-0.21** (0.0832)	-0.12 (0.0971)	-0.28*** (0.0741)	-0.33*** (0.0985)	-0.24 (0.1420)
Public health insurance coverage	-0.01 (0.0083)	0.03*** (0.0064)	0.02*** (0.0061)	0.03* (0.0158)	0.001 (0.0162)	-0.003 (0.0261)
Replacement rate of UB (60 months)	-0.06 (0.0448)	-0.12* (0.0639)	-0.13* (0.0717)	-0.10 (0.0794)	-0.04 (0.1417)	-0.01 (0.0979)
Poverty rate	-0.12*** (0.0387)	-0.09 (0.0488)	0.10 (0.0617)	0.07 (0.0704)	0.15 (0.1671)	-0.10 (0.0969)

Notes: Outcome is the **log mortality rate from alcohol use disorder** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. If figures report percent changes, coefficients can be transformed as $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 5: Dynamic effects of EPL deregulation on mortality from violence: interaction terms

Interaction term with	0y	1y	2y	3y	4y	5y
Avoidable death rate	0.08* (0.0395)	0.09 (0.0756)	0.07 (0.0762)	0.03 (0.0846)	-0.13 (0.1000)	0.04 (0.1450)
Social transfers expenditure	-0.02 (0.0612)	0.03 (0.0613)	-0.14 (0.1603)	-0.12 (0.1534)	-0.21 (0.2052)	0.07 (0.2606)
Public health insurance coverage	0.01 (0.0045)	-0.01 (0.0056)	0.01 (0.0107)	-0.01 (0.0167)	-0.03 (0.0339)	-0.06 (0.0359)
Replacement rate of UB (60 months)	0.06 (0.0684)	0.12 (0.0831)	0.07 (0.1137)	0.28** (0.0984)	0.39*** (0.1204)	0.54*** (0.1166)
Poverty rate	-0.04 (0.0928)	-0.07 (0.1088)	-0.29** (0.1043)	-0.10 (0.1484)	-0.07 (0.1439)	-0.20 (0.1843)

Notes: Outcome is the **log mortality rate from violence** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Coefficients can be transformed to percent changes using $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 6: Dynamic effects of EPL deregulation on mortality from inflammatory heart diseases: interaction terms

Interaction term with	0y	1y	2y	3y	4y	5y
Avoidable death rate	-0.004 (0.0159)	0.003 (0.0197)	0.02 (0.0175)	0.07*** (0.0203)	0.08** (0.0287)	0.07** (0.0266)
Social transfers expenditure	-0.06** (0.0234)	-0.02 (0.0317)	-0.04 (0.0539)	0.02 (0.0490)	-0.02 (0.0721)	0.02 (0.0469)
Public health insurance coverage	-0.002*** (0.0004)	0.001 (0.0010)	-0.005 (0.0027)	-0.001 (0.0043)	-0.02** (0.0098)	-0.03* (0.0177)
Replacement rate of UB (60 months)	-0.02 (0.0286)	-0.01 (0.0416)	0.01 (0.0572)	0.002 (0.0537)	0.0002 (0.0602)	-0.02 (0.0585)
Poverty rate	-0.05 (0.0318)	-0.06 (0.0603)	-0.10** (0.0427)	-0.19** (0.0658)	-0.42*** (0.0351)	-0.30*** (0.0551)

Notes: Outcome is the **log mortality rate from inflammatory heart diseases** (per 100,000 population). Entries are interaction effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Percent-change interpretation (if used in figures): $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 7: Dynamic effects of EPL deregulation on mortality from drug use disorders by age group

Age group	0y	1y	2y	3y	4y	5y
15–19	0.03 (0.1147)	-0.03 (0.1774)	-0.09 (0.2286)	-0.05 (0.1927)	-0.08 (0.1464)	-0.55*** (0.1176)
20–24	0.00 (0.1006)	0.14*** (0.0375)	0.17* (0.0949)	0.40*** (0.0700)	0.16 (0.1140)	0.36*** (0.0668)
25–29	-0.07 (0.1093)	0.07 (0.0665)	0.38*** (0.0874)	0.22** (0.0825)	0.35*** (0.0510)	0.50*** (0.1179)
30–34	0.08 (0.0935)	0.12 (0.1545)	0.31*** (0.0557)	0.16 (0.0904)	0.44*** (0.1067)	0.42*** (0.0691)
35–39	0.04 (0.0456)	0.11*** (0.0341)	0.23*** (0.0657)	0.20*** (0.0557)	0.32*** (0.0700)	0.29*** (0.0500)
40–44	0.01 (0.0614)	-0.10 (0.0824)	0.12 (0.0778)	0.17** (0.0565)	0.22*** (0.0533)	0.26*** (0.0615)
45–49	0.17** (0.0668)	0.12** (0.0550)	0.17** (0.0604)	0.11 (0.0722)	0.07 (0.0982)	0.25*** (0.0631)
50–54	-0.04 (0.1236)	0.22** (0.0915)	0.13** (0.0538)	0.19 (0.1333)	0.23*** (0.0531)	0.48*** (0.0554)
55–59	0.01 (0.1226)	0.11 (0.0861)	0.08 (0.0963)	0.15* (0.0787)	0.13* (0.0704)	0.34** (0.1245)
60–64	0.19* (0.0973)	0.24** (0.0961)	0.35*** (0.0750)	0.40** (0.1317)	0.27*** (0.0749)	0.41*** (0.1029)
65–69	0.07 (0.0961)	0.25* (0.1327)	0.12 (0.1489)	0.32*** (0.0937)	0.20*** (0.0641)	0.38*** (0.0893)
70–74	-0.30** (0.1207)	0.06 (0.1104)	-0.05 (0.0935)	-0.24*** (0.0772)	0.13 (0.1118)	-0.17 (0.1126)
75–79	-0.10 (0.1498)	0.11 (0.0986)	0.14 (0.0913)	0.06 (0.0879)	0.16 (0.1804)	0.14 (0.1384)

Notes: Outcome is the **log mortality rate from drug use disorders** (per 100,000 population). Entries are age-group-specific dynamic effects at horizons $h = 0$ to 5 years after a major EPL deregulation; standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For percent-change interpretation (used in figures if applicable), transform as $100[\exp(\hat{\beta}_h) - 1]\%$.

Table 8: ICD-10 Codes and Descriptions Grouped by Cause of Death

Cause of Death	ICD-10 Code	Description
Drug Use Disorders	F11	Mental and behavioral disorders due to use of opioids
	F12	Mental and behavioral disorders due to use of cannabinoids
	F13	Mental and behavioral disorders due to use of sedatives or hypnotics
	F14	Mental and behavioral disorders due to use of cocaine
	F15	Mental and behavioral disorders due to use of other stimulants, including caffeine
	F16	Mental and behavioral disorders due to use of hallucinogens
	F18	Mental and behavioral disorders due to use of volatile solvents
	F19	Mental and behavioral disorders due to multiple drug use and other psychoactive substances
	X41	Accidental poisoning by antiepileptic, sedative-hypnotic, antiparkinsonism and psychotropic drugs
	X42	Accidental poisoning by narcotics and hallucinogens
	X44	Accidental poisoning by other and unspecified drugs
Alcohol Use Disorders	F10	Mental and behavioral disorders due to use of alcohol
	X45	Accidental poisoning by and exposure to alcohol
Self-Inflicted Injuries	X60–X84	Intentional self-harm
	Y870	Sequelae of intentional self-harm
Violence	X85–Y09	Assault
	Y871	Sequelae of assault
Inflammatory Heart Diseases	I30	Acute pericarditis
	I31	Other diseases of pericardium
	I32	Pericarditis in diseases classified elsewhere
	I33	Acute and subacute endocarditis
	I38	Endocarditis, valve unspecified
	I40	Acute myocarditis
	I42	Cardiomyopathy

Table 9: Summary statistics

Variables	Count	Mean	SD	Min	Max
Outcome variables					
drug_mortality	486	2.561944	2.753695	0	19.09016
suicide_mortality	484	10.895560	4.007673	3.321734	26.31062
alcohol_mortality	486	2.456993	2.144824	0	10.89471
violence_mortality	486	1.140382	1.161581	0	7.333786
inflammatory_heart_disease_mortality	488	3.654734	1.323953	1.039139	8.607623
Treatment variable					
epl_flag	486	0.0411523	0.2535397	-1	1
Control variables					
avoidable_death_rate	458	220.9061	65.47542	124	515
replacement_rate_UB_60_months	440	32.85455	16.36616	0	69
gov_social_transfers_expenditure	457	11.89355	3.417891	5.023348	19.67462
nominal_GDP_per_capita_dollar	486	40205.44	15698.44	10665.07	121073.8
manufacturing_share_GDP	486	15.05076	5.52897	4.554389	34.89132
unemployment_rate	463	7.262532	4.02456	2.01395	26.49027
poverty_rate	267	10.39851	3.362702	4.545772	18.62064
gov_health_insurance_coverage	462	95.63507	15.33584	22	100

Notes: Mortality rates are per 100,000 population. `epl_flag` is an indicator of major EPL deregulation episodes (range -1 to 1 as coded in the dataset). Summary statistics computed over available observations; counts reflect non-missing entries for each variable.

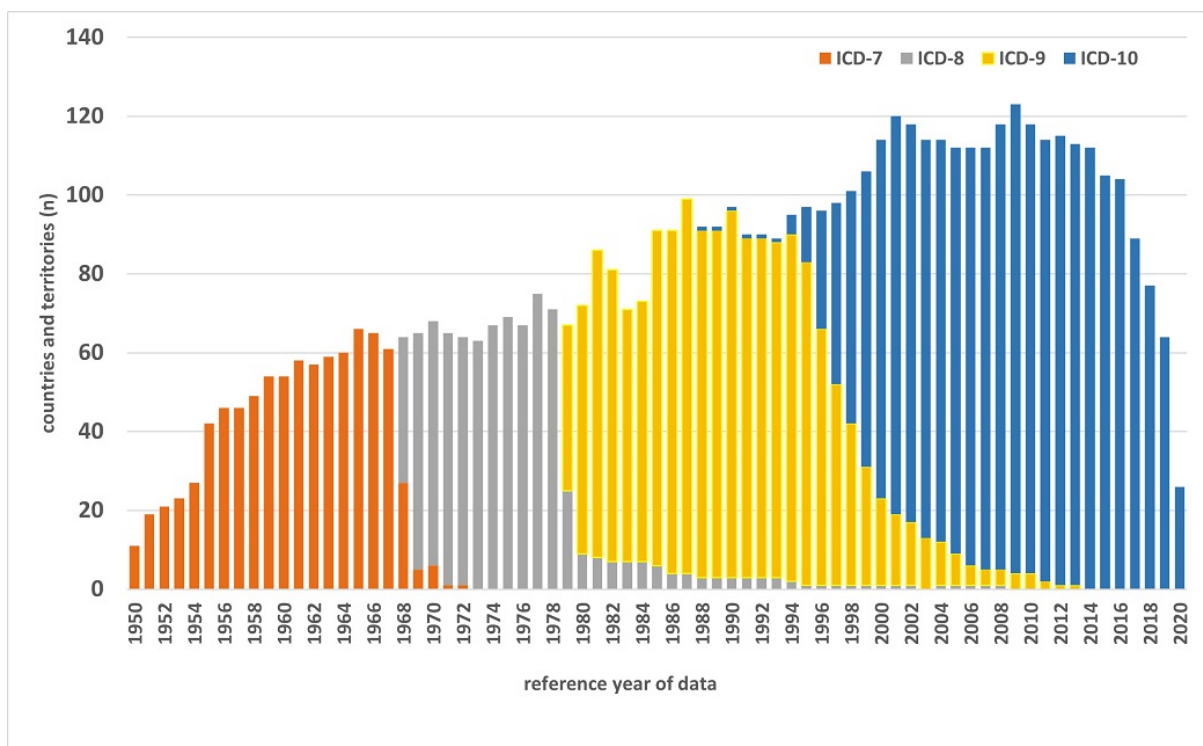


Figure 9: Number of Countries in ICD Transition